

GRIPPY: A VR Grip Controller for Combating Sarcopenia in Elderly

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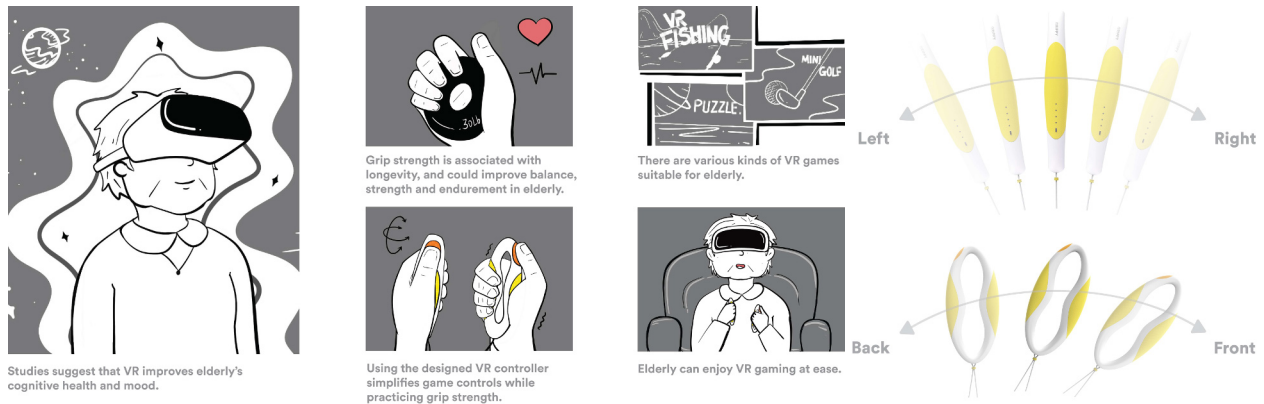


Figure 1: A conceptual VR controller that enables the elderly to exercise handgrip in VR games.

ABSTRACT

As the global population ages, sarcopenia—age-related muscle decline—demands innovative solutions. This paper introduces GRIPPY, a VR grip controller that transforms basic handgrip exercises into immersive, gamified tasks. By integrating sensor-based interaction and VR gameplay, older adults can strengthen grip while maintaining higher motivation. A preliminary test with three participants (ages 68–80) revealed key usability insights, underscoring GRIPPY's potential for enhanced rehabilitation. Future work will refine its design and validate long-term efficacy.

Index terms: Virtual Reality, Sarcopenia, Grip Strength, Elderly Rehabilitation, Immersive Exercise, Interactive Training

1 INTRODUCTION

As VR technology matures, it offers novel approaches to enhance older adults' leisure and rehabilitation activities. Sarcopenia,

defined by progressive skeletal muscle loss, poses significant risks, including frailty, falls, and reduced quality of life [2]. Handgrip exercises specifically target forearm muscle strength and are widely recommended to help manage sarcopenia [3]. However, many existing grip-strength tools are purely mechanical and repetitive, leading to limited adherence [7], [8]. VR-based interventions can provide interactive, adaptive environments that gamify exercise, thereby potentially improving motivation and neuromuscular activation [1], [5].

This paper introduces GRIPPY, a VR grip controller that converts conventional handgrip routines into immersive, gamified sessions. We discuss the system's conceptual underpinnings, detail the mechanical and sensing elements, and present results from a preliminary user test involving three older adults with varying muscle conditions. Finally, we outline directions for future research, including larger-scale clinical evaluations and extended usage guidelines.

2 NEED FOR ENGAGING HANDGRIP EXERCISES

Research consistently underscores handgrip exercises as a primary modality for addressing sarcopenia in older adults [2]. Gradually increasing muscle load promotes hypertrophy and optimises neuromuscular function [3]. Forearm muscles are pivotal for grip strength, and targeted exercises can help older adults maintain dexterity for daily tasks, such as lifting objects. Large-scale analyses, including the Korea National Health and Nutrition Examination Surveys, affirm that strength-based activities like handgrip training significantly reduce sarcopenia rates [6].

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Despite proven benefits, many existing handgrip devices are purely mechanical and lack engaging features, reducing long-term motivation [7], [8]. While essential devices are accessible [2], their repetitive nature often results in limited adherence. VR technologies offer interactive, adaptive environments that can gamify the exercise experience, potentially increasing adherence and enhancing neuromuscular activation [1], [5]. By merging sensor-based data acquisition with immersive gameplay, we can transform handgrip exercises into a more compelling, user-centric activity for entertainment and rehabilitation.

3 INTRODUCING VR-BASED GRIP CONTROLLER

Building on these insights, we introduce GRIPPY (see Figure 1), a grip-style VR controller that combines a mechanical handgrip with digital sensor interfaces to elevate the user experience:

3.1 Elastic Grip Ring and Flex Sensor

GRIPPY employs a deformable, elastic ring. As the user applies force, the ring's shape changes and an embedded flex sensor measures the resulting grip strength in real-time.

3.2 Control Circuit and Wireless Communication

An internal control circuit integrates a microcontroller, power supply, and wireless module, enabling seamless data transmission between the handgrip and an external VR environment.

3.3 Enhanced User Engagement

By blending game-based feedback loops with visually rich VR scenes, GRIPPY surpasses the monotony of standard grip trainers. Adjustable force thresholds let users customise difficulty to match their capabilities, creating a highly motivating training experience.

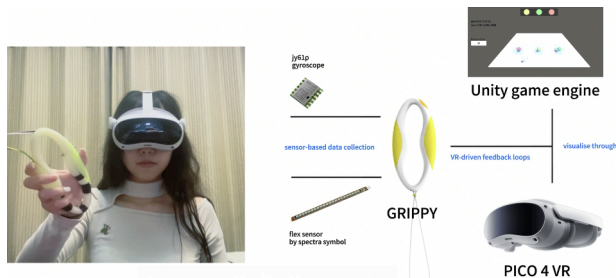


Figure 2: Setup of a functional prototype.

4 PROTOTYPE IMPLEMENTATION

We built a functional prototype (see Figure 2) using an Arduino microcontroller, a flex sensor, and a gyroscope to track orientation. This setup communicates with a Unity scene that displays objects in corresponding positions. Users manipulate the virtual environment by varying their grip force, and the game updates in real-time to reflect successful pickups, releases, and positioning tasks.

5 PRELIMINARY USER TEST

We tested GRIPPY with three older adults (ages 68–80, varying muscle conditions) under low, medium, and high force thresholds. Two participants had no VR experience. All found the low threshold easiest, though two struggled with direction control at higher forces due to hand shakiness. One desired guidance on ideal thresholds and usage duration, noting that prolonged gripping caused fatigue. Another reported occasional mismatch between perceived and actual pickup positions, suggesting a need for clearer visual cues. Despite these challenges, participants appreciated the

gamified approach, emphasizing that precise control, adaptive difficulty, and clear feedback are critical for improving GRIPPY's usability and long-term rehabilitation value.

6 CONCLUSION AND FUTURE WORK

GRIPPY shows promise in transforming repetitive handgrip exercises into engaging, VR-based interactions for older adults. By mapping grip force to immersive tasks, it can enhance motivation and adherence—vital factors for effective sarcopenia management.

Our preliminary test highlights the importance of balancing low and high force thresholds. While lower thresholds ease adoption, higher ones pose control challenges but offer greater training potential. Shakiness at higher forces indicates a need for refined input mapping and clearer visual or haptic cues, ensuring smoother orientation and interaction. Additionally, participants sought more explicit guidelines on threshold selection and session duration to optimise safety and overall effectiveness.

Moving forward, we will refine ergonomics by exploring alternative designs and materials that reduce user fatigue, and develop adaptive protocols that dynamically adjust thresholds in response to real-time performance. Moreover, longitudinal studies will validate GRIPPY's effects on grip strength and functional outcomes, cementing its clinical value. Finally, we plan to integrate safety monitoring features like muscle fatigue or heart rate tracking to offer personalised training limits and support healthy ageing.

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